MIL-F-50070C
30 SEPTEMBER 1964
SUPERSEDING
MIL-F-50070B(CBRA)
13 SEPTEMBER 1962

MILITARY SPECIFICATION

FACEBLANK, C17

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a protective mask faceblank molded from a 65-percent natural rubber compound.

1.2 Classification. The faceblank shall be of the following sizes, as specified (see 6.2):

Small

Medium

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-601 — Rubber; Sampling and Testing.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

E5-2-966 — Faceblank, Small.

E5-2-968 — Faceblank, Medium.

PUBLICATIONS

EDGEWOOD ARSENAL DIRECTIVE

No. 618 — Ozone Resistance Test of Natural Rubber Material.

(Copies of specifications, standards, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations

FSC 4240

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(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials.

3.1.5 *Coloring materials.* The rubber compound shall be nonblooming, except for a slight wax-type bloom, and shall contain no less than 65 percent, by volume, of new natural rubber. The compound shall be free from lead or mineral sulfates.

3.1.2 *Antioxidants and accelerators.* The rubber compound shall contain only approved antioxidants and accelerators (see 3.1.6).

3.1.3 *Antiozonide.* The rubber compound shall contain an antiozonide.

3.1.4 *Copper and manganese inhibitor.* The rubber compound shall contain a copper and manganese inhibitor.

3.15 *Coloring materials.* The rubber compound shall be colored by carbon black as required.

3.1.6 *Toxicity.* The faceblank shall not cause undue primary skin irritation or sensitization of the skin when assembled and worn as a component of protective masks. Objective evidence shall be submitted in accordance with 4.2.2.2. One hundred and twenty-five masks and thirty test slabs, having the same formulation and equivalent cure (see 6.3) as will be used during regular production, shall be submitted to the Government for testing as specified in 4.7.3, except when the formulation has been approved prior to preproduction. Thirty additional test slabs shall be submitted for approval under the conditions cited above whenever a major change is made (see 6.4). The supplier shall not make a major change in rubber formulation or the curing process after receiving approval of a formulation from the Government.

3.2 Manufacture. The faceblank shall be manufactured in one piece by the conventional process of molding under heat and pressure. Each size specified in the contract or order shall be manufactured in accordance with the applicable drawing listed in 2.1 in molds approved by the Government.

3.2.1 *Repair of flaws.* Faceblanks with small flaws (major or minor defects) may be repaired and accepted provided the repair is vulcanized with material of the same approved formulation as the faceblank and the repaired area meets the requirements of physical properties of 3.3. The appearance of the repair shall be such that the color and shape blend into the surrounding area with no break in surface continuity and no feather edges.

3.2.2 *Test slabs.* Test slabs 6 by 6 inches, by 0.075 $\pm$ 0.01 inch shall be made from stock representative of the formulation used for the preproduction sample and from each sublot during regular production of faceblanks. The slabs shall have a cure equivalent to that of the regular production faceblank (see 6.3).

3.3 Physical properties. Physical properties shall conform to the requirements shown in table I when tested as specified therein.

3.4 Preproduction sample. Prior to the start of regular production, 30 (20 trimmed and 10 untrimmed) sample faceblanks from each mold cavity and 16 test slabs shall be produced using the same methods, materials, equipment, and rubber formulation as will be used during regular production. Any change in formulation, operating conditions, or molds shall require a new preproduction sample.

3.5 Critical areas. The peripheral sealing area contacting the wearers' skin, the inner and outer surfaces of the element nozzles, the voicemitter outlet valve, and the lens sealing surface shall be considered as critical areas of the faceblank.

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TABLE I. *Physical properties*

Property	Unaged	After heat aging	Test paragraph
Average tensile strength, psi, min.	3000	2550	4.7.2.1
Average ultimate elongation, %, min.	600		4.7.2.1
Percent increase, max.		20	
Percent decrease, max.		25	
Average tensile stress at 200% elongation, psi	250 to 450	250 to 450	4.7.2.1
Individuals, psi, min.	240	240	
Average tear resistance, lbs/in., min.			4.7.2.1
Die A or B	300	210	
Die C	225	165	
Average tension set at 75% ultimate elongation, %, max.			4.7.2.1
Average hardness (Shore "A" or "A2")	47 ± 3		4.7.2.1
Ozone resistance, min.	80		4.7.2.2
Flexibility			4.7.2.3
Units at 75° ± 5°F., min.	7.5		
Units retained at —25° ± 5°F., min.	4.8		
Puncture resistance, psi, min.	9.5		4.7.2.4

3.6 Workmanship. The faceblank inner and outer surfaces shall be smooth and free from imperfections such as pits, bumps, porous areas, cracks, flash, abrasions, checks, blisters, or foreign matter. The rubber shall be compact and free from holes and bubbles. Ridges of mold-parting lines on outer surfaces shall not be considered as flash when closely trimmed. Rips, tears, or punctures shall be considered as damage. Workmanship defects of a magnitude greater than the referee standards established by the Government for major A, major, and minor defects (see 6.8) shall be categorized in the appropriate defect class.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier's responsibility.* Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.1.2 *Government's responsiblity.* The Government will be responsible for the performance of test in 4.7.3. Samples shall be forwarded to the laboratory designated by the contracting officer.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.*

4.2.2.1 *Materials.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2.2.2 *Toxicity.* The supplier shall submit to the contracting officer within 30 days after date of award of a contract, at no additional cost to the Government, a complete list of the ingredients to be used in the rubber formulation, the percentage of each, the cure time and temperature to be used, and the tolerances of the above that will be permitted, as well as the mold lubricants, dusting powders, or wet sprays to be used. The supplier shall also submit with the above list the physical test results and complete evidence substantiating his claim, to the satisfaction of the Government, that the listed components will be nonirritating or nonsensitizing to the wearer of the end item involved. This evidence may be in the form of a complete description of the programs that the supplier has carried out, such as patch testing or use testing. He may submit evidence that the proposed formulation is essentially the same as that used in his commercial products, and include information relative to the extent and length of time of his commercial experience with the formulation. Other evidence may include references to technical or other scientific publications, laboratory reports, opinions or statements of recognized authorities or agencies in the field of dermatology, or other pertinent data. The Surgeon General will act as advisor in evaluating this evidence for determining if the components are acceptable (see 4.7.3).

4.2.3 *Inspection substation.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL–STD–105 are applicable to each substation and to the end item.

4.3 Preproduction sample inspection.

4.3.1 *Examination.* Fifteen of the sample faceblanks shall be inspected for all requirements of the drawings and this specification.

4.3.2 *Tests.* Fifteen faceblanks and 16 test slabs, produced in accordance with 3.4, shall be tested in accordance with 4.7.2. Faceblank material, produced in accordance with 3.1.6, shall be tested in accordance with 4.7.3, except where the toxicity formulation has been approved prior to preproduction.

4.3.3 *Acceptance/rejection criteria.* The preproduction sample shall meet the examination and tests specified by 4.3.1 and 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Lotting. A regular production lot shall consist of the faceblanks of one size produced in sequence during one month as indicated by the dots in the medallion on the faceblank. A sublot shall consist of the faceblanks produced from one batch of material (see 6.5). A new lot shall be established following the occurrence of any event affecting the homogeneity of the lot or use of an alternative approved formulation.

4.5 Sampling.

4.5.1 *For examination.* Sampling shall be conducted in accordance with MIL–STD–105.

4.5.2 *For destructive tests.* Fifteen faceblanks and 16 test slabs shall be submitted with each sublot for testing.

4.6 Inspection provisions.

4.6.1 *For examination and nondestructive tests.* The sample faceblanks from the regular production lot shall be examined in accordance with the classification of defects and with MIL–STD–105.

4.6.2 *For destructive tests.* Sample faceblanks and test slabs shall be tested in ac-

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cordance with 4.7.2. The acceptance number is zero.

4.6.3 *Classification of defects.*

4.6.3.1 *Faceblank.* See applicable drawing for size.

Categories	*Defects*
Critical:	None defined.
Major:	AQL 1.0 percent defective.
101	Rubber thickness incorrect.
102	Porosity or nonfills.
103	Deformation or permanent set.
104	Flow cracks which enlarge on probing.
105	Marking incorrect.
106	Workmanship (other than Major A 150, see 3.6).
107	Mismatched bag mandrels.
108	Split at register.
Major A:	AQL 0.065 percent defective.
150	Imperfections or damage in critical areas.
Minor:	AQL 2.5 percent defective.
201	Excess flash or improper trim.
202	Marking missing or illegible.
203	Access flap holes improperly located.
204	Incorrect size.

4.7 **Tests.**

4.7.1 *Test specimens.* Two specimens, one for each cheek, shall be cut from the flat cheek areas of each of 15 sample faceblanks. The specimens shall not be buffed or stretched before testing. Ten specimens, five from each cheek, shall be tested for tensile strength, ultimate elongation, and tensile stress before aging and 10 specimens tested after aging. Six specimens, three from each cheek, shall be tested for tension set before aging. Two specimens for tear resistance shall be cut from the forehead section of each of 10 of the above 15 sample faceblanks. Ten specimens, five from each cheek, shall be tested before aging and 10 tested after aging. Enough 1 by 1-inch squares shall be cut from cheek and forehead sections to build up five specimens for the hardness test, each specimen containing enough layers to be no less than $\frac{3}{16}$ inch thick. The remaining faceblanks shall be used for retest (4.8). Twelve specimens, two from each of six test slabs, shall be taken for ozone resistance tests. Eight specimens, two from each of four test slabs, shall be taken for flexibility testing. Six specimens, one from each of six test slabs, shall be taken for puncture resistance tests. The test specimens selected for heat aging shall be aged in an air oven at a temperature of 158° ± 1.8°F. for a continuous period of 168 ± 1 hours or at a temperature of 212° ± 1.8°F. for 24 ± 1 continuous hours, as specified in FED-STD-601 (see 6.6).

4.7.2 *Physical properties tests.*

4.7.2.1 *Tensile strength, ultimate elongation, tensile stress, tear resistance, tension set, and hardness.* Tensile strength, ultimate elongation, tensile stress, tear resistance, tension set, and hardness shall be determined as specified in FED-STD-601. The angle-type or crescent-type die shall be used for tear resistance testing.

4.7.2.2 *Ozone resistance.* Ozone resistance shall be determined in accordance with Directive No. 618.

4.7.2.3 *Flexibility.* Flexibility shall be determined using Edgewood Arsenal Q37 Natural Rubber Flexibility Tester. Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 75° ± 5°F. and four specimens shall be tested at —25° ± 5°F. after being conditioned for 72 hours at that temperature.

4.7.2.4 *Puncture resistance.* Resistance to hydrostatic pressure shall be determined using Edgewood Arsenal Q38 Natural Rubber Puncture Tester.

4.7.3 *Toxicity.* All masks and test slabs submitted to the Government for testing are subject to evaluation from the toxicity standpoint by the Office of the Surgeon General.

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Such tests as are deemed necessary of any protective mask or test slab will be carried out in accordance with direction of the Surgeon General.

5. PREPARATION FOR DELIVERY (For repair parts packaging see 6.7).

5.1 Packaging, level C. Faceblanks of one size shall be packaged in such a manner as to prevent any deformation or damage during shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Faceblanks of one size, packaged as specified in 5.1, shall be packed in such manner as to insure carrier acceptance and safe delivery at first destination for immediate use. Containers used shall comply with the Uniform Freight Classification rules or regulations governing the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129. The following additional marking shall appear on all unit packages and shipping containers.

(a) Faceblank size.
(b) Composition of faceblank and formulation code.
(c) Nomenclature of mask to which faceblank pertains.
(d) Date of cure.

6. NOTES

6.1 Intended use. The faceblank covered by this specification is intended for use in the manufacture of ABC-M17 Field Protective Masks.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Size of faceblank.
(c) Applicable drawings.

6.3 Equivalent cure. Slabs and faceblanks of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent.
(b) Their average tensile-stress values at any given elongation (other than ultimate elongation) do not vary more than $\pm$ 2.5 percent.
(c) Their curing times can be equated in the following equation:

$$\frac{T_1 - T_2}{10}$$

$$t_2 = t_1 \times 1.6$$

where: t_1 = minutes of cure at temperature T_1,

t_2 = minutes of cure at temperature T_2,

T_1 = temperature (°F.) of cure for time t_1, and

T_2 = temperature (°F.) of cure for time t_2.

For convenience, the nomograph in figure 1 may be used instead of the mathematical computation time-temperature relationship in determining equivalent cure.

6.4 Major change. The term applies to any change in formulae, processing, or curing of currently approved rubber stocks which induce toxic characteristics and cause casualties. The major change should be considered effected when one or more of the following conditions occur:

(a) Components of formulae are varied in excess of the limitations stated in the manufacturer's tolerance (see 4.2.2.2).

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EQUIVALENT CURE TIME

FOR

NATURAL RUBBER

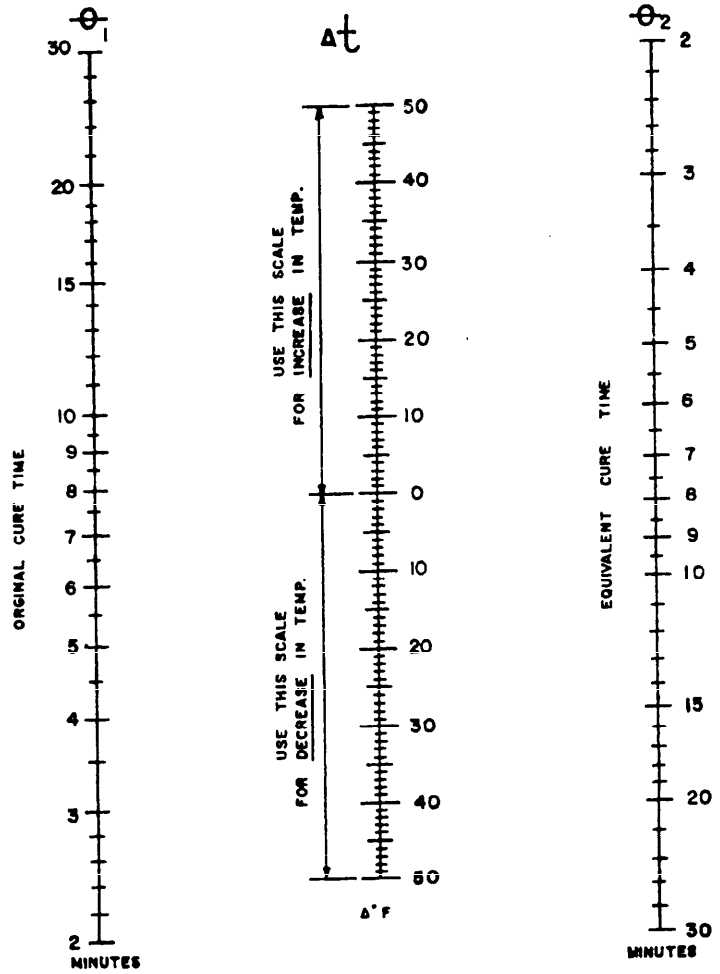


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

(b) The addition or deletion of one or more components of formulae.

(c) Substitutes for one or more components of formula without advance approval of the Government.

(d) Deviation from the submitted approved factory mold cure exceeding + 5°F. or — 10°F., even though equivalent cure is attained.

(e) Use of mold lubricants, dusting

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powders, or wet sprays not used in preparation of cure of the original samples submitted for toxicity approval.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 Aging. Aging at 212° ± 1.8°F. for a continuous period of 24 hours is slightly more severe on rubber than aging at 158° ± 1.8°F. for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 158° ± 1.8°F. for 168 consecutive hours.

6.7 Repair parts packaging. When repair parts packaging and packing is required for items authorized for stocking as repair parts, the repair parts packaging data sheet covering the C17 Faceblank should be used. (Copies of repair parts packaging data sheets may be obtained from the Director of Engineering & Industrial Services, ATTN: Chief, Design Branch, U.S. Army Edgewood Arsenal, Md. 21010)

6.8 Referee standards. During preproduction and subject to later revision during production, the Government will establish referee standards to define the type and category (Major, Minor, and Major A) of each defect. This classification will be based upon the defect magnitude and potential deleterious effect on intended use.

6.9 International interest. Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:

Army—MU

Navy—SH

Air Force—84

International interest (see 6.9)

Review activities:

Army—MD

Navy—SH

User activities:

Navy—MC

Air Force—84

Preparing activity:

Army—MU(EA)

Project No. 4240-0062

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